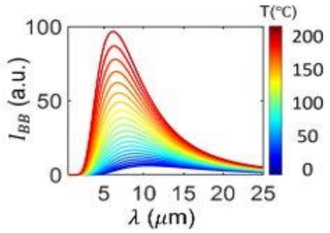
Energy and Coherence Amplification of Blackbody Radiation in 1D Photonic Time-Crystals

Luis Cortes-Herrera[1], **Naren Ganesh**[1], Jack Hasty[1], Yuzhe Xiao[1]

[1]Department of Physics, University of North Texas, Denton, Texas 76205, USA

Introduction

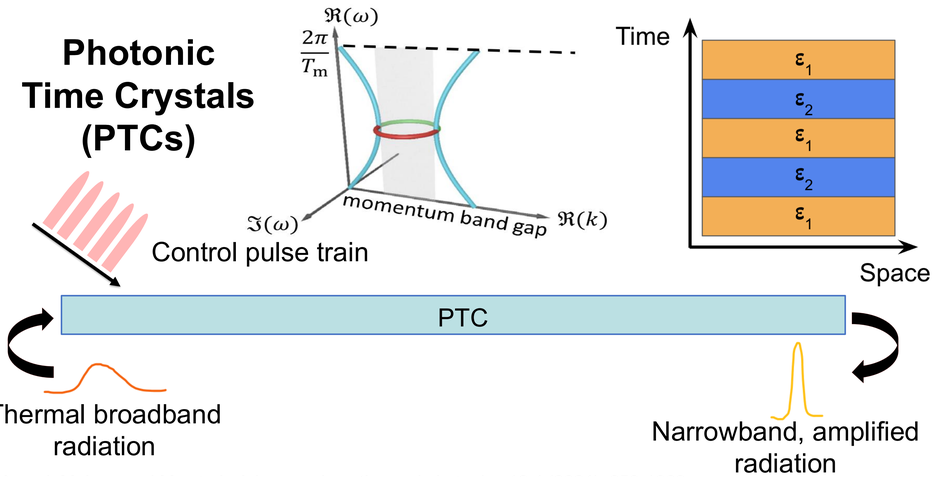
Thermal Radiation



Thermal radiation is

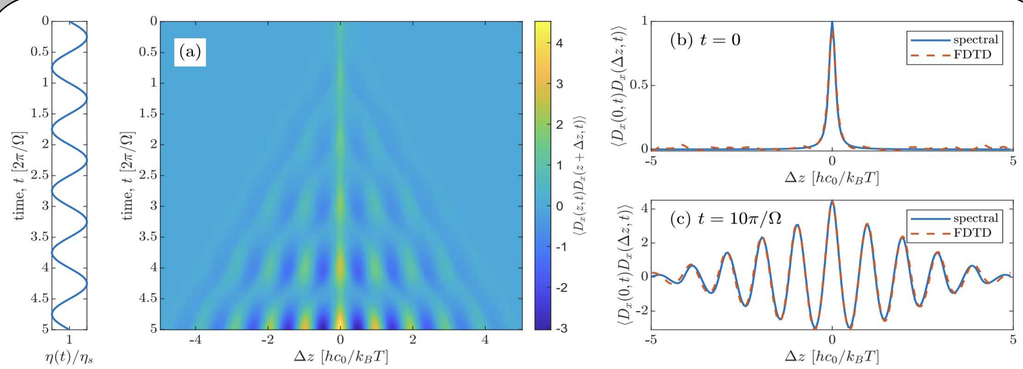
- Broadband
- Stochastic
- Temperature dependent

Photonic Time Crystals (PTCs)



[1] Asgari, Mohammad M., et al. Advances in optics and photonics 16.4 (2024): 958-1063.

Methodology



Refractive index modulation

(a) Evolution of spatial autocorrelations of the initially thermal field $D_x(z,t)$

Panels (b,c) validate methodology of spatial-spectral theory (solid) with stochastic FDTD simulations (dashed).

Initial Thermal State → **Time Evolution in PTC** → **Time-varying Fields**

Thermal equilibrium Gaussian random fields

$\langle a_k(0)a_k^*(0)\rangle \propto \hbar\omega_k / (e^{(\hbar\omega_k/k_BT)} - 1)$

Selective amplification governed by band structure

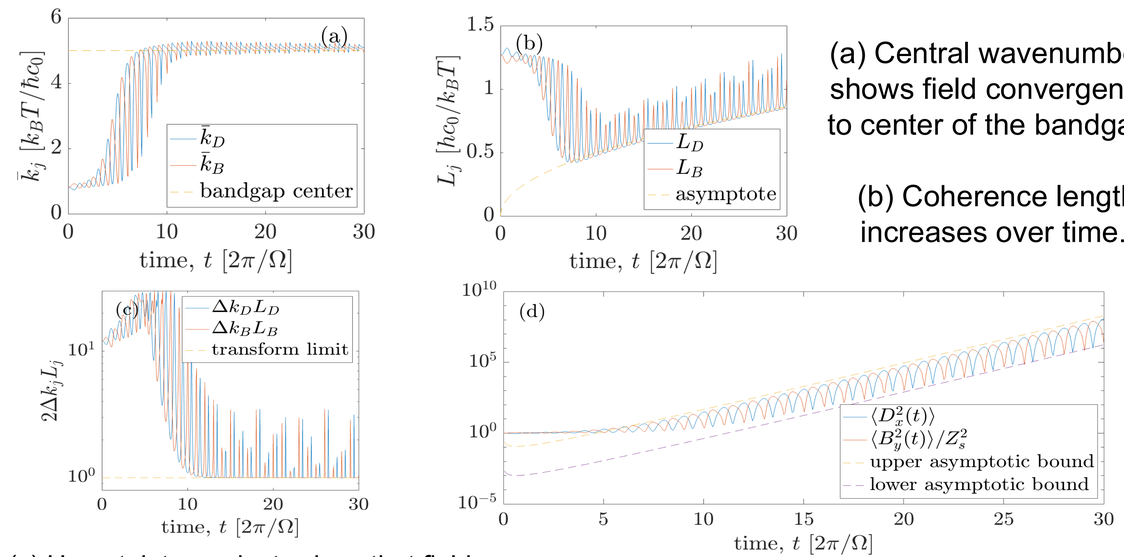
$i\, d/dt\, u_k(t) = H_k(t)\, u_k(t)$

Synthesize correlations by Spatial Fourier Transform

$\langle D_x(z,t)\, D_x(0,t)\rangle = \int W_D(k,t)\, e^{ikz}\, dk$

Results

Long-time asymptotics



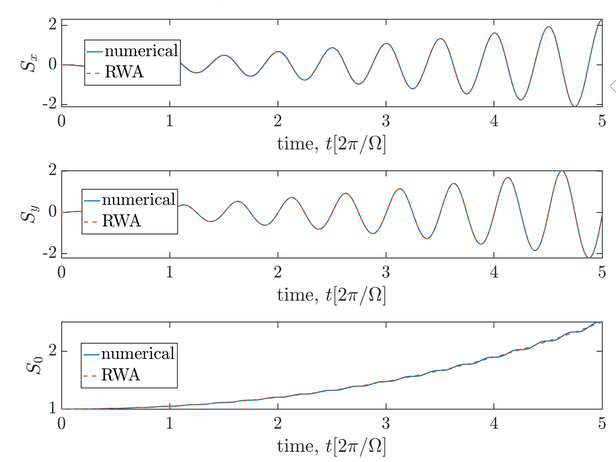
(a) Central wavenumbers shows field convergence to center of the bandgap.

(b) Coherence length increases over time.

(c) Uncertainty products show that field converges to Gaussian FT limit.

(d) Second moments show increase in electromagnetic energy.

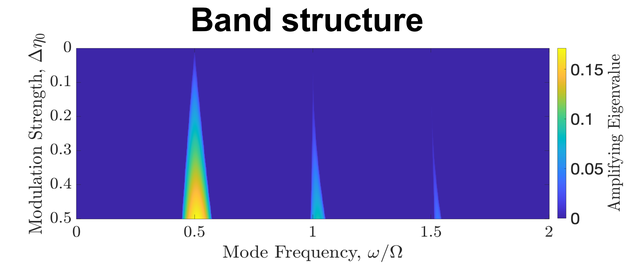
Rotating Wave Approximation



Long-time asymptotics emerge from the bandgap center where the rotating wave approximation is accurate.

Time evolution of the Bloch parameters S_x, S_y, and S_0 at the center of the momentum bandgap.

Band structure



Magnitude of the amplifying eigenvalue against mode frequency and modulation strength.

Conclusion

- Spatial spectrum becomes periodically Gaussian (FT limit).
- Determine asymptotics based on band structure.
- Employ rotating wave approximation to study evolution at bandgap center.

Acknowledgement

This project is supported by DARPA YFA program (Grant No. D23AP0018900) and NSF LEAPS-MPS program (Award Number 2418002).